

Task Descriptions as Bayesian Priors: How Many Examples Is an Instruction Worth?

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Abstract

A prompt to a language model usually contains a task description and some examples. Theories of in-context learning (ICL) explain what the examples do but not what the description adds, or how the two trade off. We treat ICL as Bayesian inference over a latent task, so that examples are data and a description sets the prior, and we measure a description’s worth as its *effective sample size*: the number of examples that give a Bayes-optimal learner the same prediction regret. In a testbed where the optimal learner is exact, in-context linear regression, we compute this worth as a function of a description’s *precision* (how narrowly it constrains the task) and *reliability* (how often it is correct). Three findings. First, a description is worth more examples for a single prediction than for a long session with feedback, by up to $8.5\times$ on our grid; the long-horizon worth equals the description’s information content, and the gap is geometric: a description narrows every direction of the task while examples narrow only the directions they cover. Second, reliability caps the worth of precision: a description that is wrong one time in ten gains little from being made fifty times more precise, whereas a reliable one gains sixfold. Third, for precise but unreliable descriptions the ordering reverses, and the description is worth more over a long session than for a single prediction, because feedback lets the learner check it. [Fourth, on the trained networks, once obtained.] All claims concern Bayes-optimal predictors and small meta-trained sequence models, not large language models.

1 Introduction

Theories of in-context learning have mostly studied prompts made only of input–output examples (Garg et al., 2022; Akyürek et al., 2023; Xie et al., 2022; Genewein et al., 2025). Real prompts usually contain more: a natural-language task description together with the examples (Brown et al., 2020). Example-only theories cannot say what the description adds, or how it trades off against the examples, and the empirical record points both ways. A prompt can be worth hundreds of labelled examples (Le Scao and Rush, 2021; Honda et al., 2025), yet models also discount instructions: enough examples override an explicitly stated bias (Gupta et al., 2025), and misleading instructions can work about as well as good ones (Webson and Pavlick, 2022). Theory has begun to include instructions (Huang and Ge, 2025; Lin et al., 2025; Tong et al., 2026) but does not yet say how many examples an instruction is worth.

We give a Bayesian answer. If ICL is Bayesian inference over a latent task (Xie et al., 2022), examples are data and the description sets the prior. Bayesian statistics already measures what a prior is worth in data: its effective sample size, the number of observations the prior is equivalent to (Morita et al., 2008). A prior’s worth depends on how concentrated it is and on whether it is right (Evans and Moshonov, 2006), and the same two properties describe an instruction:

- **Precision:** how narrowly the description constrains the space of tasks.
- **Reliability:** how often the description is correct.

We compute a description’s worth as a function of both, in a setting where the Bayes-optimal learner is exact: in-context linear regression (Garg et al., 2022; Akyürek et al., 2023) with a Gaussian prior on the task whose mean and variance a description sets, and a robust mixture prior (Schmidli et al., 2014) when the description may be wrong. We define worth through prediction, as the number of examples whose cumulative regret over the next N predictions equals the description’s, and we ask how it depends on the horizon N .

Contributions.

1. A definition of a description’s worth in examples through prediction regret at a chosen horizon (§3). Prediction-based effective sample sizes were left open in the statistics literature (Reimherr et al., 2021; Neuenschwander et al., 2020); ours is exact in the linear-Gaussian setting.
2. For reliable descriptions, the single-prediction worth exceeds the long-horizon worth (§4). The long-horizon worth is the examples carrying the same information as the description, a consequence of Clarke and Barron (1990); the single-prediction worth is larger because a description shrinks uncertainty in every direction and examples do not. Both have closed forms in the proportional limit through the Marchenko–Pastur law (Tulino and Verdú, 2004). The ratio grows with dimension and signal-to-noise ratio and reaches 8.5 on our grid.
3. For unreliable descriptions, the regret splits into a Gaussian term and an identification term (§5). Reliability caps the worth of precision for a single prediction; over a long horizon the worth is unbounded in precision, so the horizon ordering reverses for precise, unreliable descriptions. Mis-set trust costs $\text{KL}(p \parallel q)$ nats over a long horizon, and a wrong description is abandoned after one to three examples.
4. [Meta-trained Transformers: whether their worth in examples and their trust match the Bayes-optimal learner (§6). Results pending.]

2 Related work

Bayesian views of ICL. Xie et al. (2022) explain ICL as implicit Bayesian inference over a latent concept. In linear regression, transformers trained on Gaussian tasks track the Bayes-optimal (ridge) predictor (Akyürek et al., 2023; Raventós et al., 2023; Panwar et al., 2024), and Lu et al. (2025) give the proportional-limit error of linear attention through random matrix theory. Jeon et al. (2024) decompose the average cumulative log loss of ICL by mutual information; Genewein et al. (2025) score small meta-trained networks against the Bayes-optimal predictor by cumulative log-loss regret, which we adopt. None of these prices a prior in examples or compares horizons.

Instructions in ICL theory. Huang and Ge (2025) add a task descriptor to in-context linear regression, the mean of the inputs, and show that linear attention uses it to centre the inputs. Their descriptor carries no information about the mapping, so a Bayes-optimal learner values it at zero examples; ours describes the weights. Lin et al. (2025) give a hypothesis class as a prefix and observe that it helps at a fixed number of examples. Tong et al. (2026) show that a single incorrect instruction is overridden at an exponential rate in the number of demonstrations, without a reliability parameter. Lin and Lee (2024) study a Gaussian-mixture task prior, whose posterior is the same conjugate computation as ours, and show that a misleading prior hurts early and washes out. Zhu et al. (2026) train transformers on a prefix of earlier datasets and find Bayes-optimal use of it. Empirical exchange rates exist for fine-tuning (Le Scao and Rush, 2021; Cook et al., 2025) and for distilled many-shot prompts (Honda et al., 2025), not for ICL with a Bayes-optimal benchmark.

Prior effective sample size. Morita et al. (2008) define a prior’s ESS by matching its curvature to that of a posterior from a flat prior; Reimherr et al. (2021) define it through prior–data discrepancy and name a prediction-based version as an open problem; Neuenschwander et al. (2020) require consistency of expected posterior ESS and treat one-parameter models. Clarke (1996) and Lin et al. (2007) define an information-theoretic prior sample size by relative entropy. Robust mixture priors (Schmidli et al., 2014) and their ESS under conflict (Wiesenfarth and Calderazzo, 2020) come from clinical trials with historical controls. Closest to our unreliable case, Reznik (2026) derive the optimal exponent of a power prior under predictive log loss and cap the borrowed sample size by the discrepancy; that ESS is the exponent times the historical sample size, does not depend on the horizon, and has no mixture. Ours is defined by matching regret against examples, depends on the horizon, and treats a two-component mixture.

3 Setting

Task. A task is a weight vector $w \in \mathbb{R}^d$. An example is an input $x \sim \mathcal{N}(0, I_d)$ and an answer $y = w^\top x + \epsilon$, $\epsilon \sim \mathcal{N}(0, \sigma_y^2)$. Without a description, $w \sim \mathcal{N}(0, s_0^2 I_d)$. We set $\sigma_y = 1$ and write $a_0 = s_0^2 / \sigma_y^2$ for the base signal-to-noise ratio.

Description. A description is a vector $m \in \mathbb{R}^d$ and a variance $s_\ell^2 = r s_0^2$, $r \in (0, 1)$: if it is correct, $w \sim \mathcal{N}(m, s_\ell^2 I_d)$. Smaller r is a more precise description. We draw $m \sim \mathcal{N}(0, (1-r)s_0^2 I_d)$, so that w has the same marginal law with or without a description; this is the hierarchical prior of the proposal, coarse descriptions naming a task family and precise ones nearly stating the rule. A description is correct with probability p and otherwise describes an unrelated task, $w \sim \mathcal{N}(0, s_0^2 I_d)$ independently of m . The Bayes-optimal prior is the robust mixture $p\mathcal{N}(m, s_\ell^2 I) + (1-p)\mathcal{N}(0, s_0^2 I)$ (Schmidli et al., 2014). A description is idealised as a vector, not text: the learner must learn how much to weigh it, not how to parse it.

Scoring. The learner predicts a distribution for each of the next N answers, seeing each true answer before the next prediction. Its regret R_N is its cumulative log loss minus that of an oracle that knows w , in nats (Genewein et al., 2025). $N = 1$ is a single question; large N is a session with feedback.

Definition 1 (Worth in examples). Let R_N^{desc} be the regret of the learner with the description and no examples, and $R_N^{\text{ex}}(n)$ the regret of the learner with n examples and no description, both averaged over tasks and inputs. The description’s effective sample size at horizon N is the n (interpolated between integers) at which $R_N^{\text{ex}}(n) = R_N^{\text{desc}}$.

The comparison learner uses the base prior, so it is the same Bayes-optimal learner with less information. The definition follows Reimherr et al. (2021) in matching a prior against observations and differs in matching prediction regret at a horizon rather than a posterior functional.

4 Reliable descriptions: worth depends on the horizon

For $p = 1$ everything reduces to one function. Under a Gaussian prior with signal-to-noise ratio a , the posterior covariance after inputs $X_k \in \mathbb{R}^{k \times d}$ is $\sigma_y^2(I/a + X_k^\top X_k)^{-1}$ and does not depend on m or on the answers, and log-loss regret obeys the chain rule, so with

$$G_a(k) = \frac{1}{2} \mathbb{E} \log \det(I + a X_k^\top X_k), \quad (1)$$

the description’s regret over horizon N is $G_{a_\ell}(N)$ with $a_\ell = r a_0$, and the regret after n examples is $G_{a_0}(n + N) - G_{a_0}(n)$.

Proposition 1 (Long horizon). As $N \rightarrow \infty$, $\text{ESS}_N \rightarrow n^*$, the number of examples whose expected information gain equals the description’s: $G_{a_0}(n^*) = \frac{d}{2} \log(1/r)$.

This follows from the redundancy asymptotics of Clarke and Barron (1990): cumulative log-loss regret is the mutual information between the answers and w , and both learners’ regrets grow as $\frac{d}{2} \log N$ with a constant offset equal to the information each lacked at the start. In the long run a description is worth exactly the uncertainty it removes.

Proposition 2 (Single query). $\text{ESS}_1 \geq \text{ESS}_\infty$, with equality in $d = 1$ up to the variability of single examples, and exactly under balanced designs $X^\top X \propto I$.

The single-query regret is $\frac{1}{2} \mathbb{E} \log(1 + x^\top \Sigma x)$ for posterior covariance Σ , which depends on the trace of Σ (A-optimality), while the information depends on its log-determinant (D-optimality). A description shrinks every eigenvalue by the same factor; $n < d$ random examples shrink n directions almost to zero and leave the rest untouched. By the AM–GM inequality the isotropic shrinkage gives lower trace at equal log-determinant, so it takes more examples to match the description for a single prediction than for information. The inequality is classical; the contribution is its size, which we now quantify. At high signal-to-noise ratio and $n < d$, $\text{ESS}_1 \approx d(1-r)$: a description that removes a fraction $1-r$ of the prior variance is worth that fraction of d examples.

Proposition 3 (Proportional limit). As $d, n \rightarrow \infty$ with $n/d = \gamma$ and $d a_0 = \rho$ fixed, ESS_1/d solves $\eta(\gamma) = r$ and ESS_∞/d solves $\mathcal{V}(\gamma) = \log(1/r)$, where η and $\mathcal{V}$ are the η - and Shannon transforms of the Marchenko–Pastur law (Tulino and Verdú, 2004, eqs. 2.52, 2.62).

These match the simulation to within about 2% at $d = 64$.

Table 1 gives the grid. The ratio grows with dimension and signal-to-noise ratio and falls with precision: across $d \in \{1, \dots, 64\}$, $a_0 \in \{1, 10, 100\}$, $r \in \{0.9, \dots, 0.01\}$ it ranges from 1.01 to 8.5, the largest at $d = 64$, $a_0 = 100$, $r = 0.9$. In $d = 1$ it stays between 1.01 and 1.35; under balanced designs it is exactly 1. **[Figure: ESS_N/d against N for several d , with the two limits.]**

Table 1: Worth of a reliable description in examples, single query versus long horizon. $a_0 = 10$; three seeds; standard deviations across seeds below 2% of the value. Regret in log loss.

d	r	ESS ₁	ESS _∞	ratio
1	0.5	0.50	0.40	1.26
4	0.5	1.77	0.80	2.21
16	0.5	7.62	2.23	3.42
64	0.5	31.6	6.93	4.56
16	0.05	16.9	10.4	1.63
64	0.05	62.1	31.1	2.00
64	0.01	73.5	50.2	1.46

Table 2: Worth of an unreliable description, calibrated trust ($q = p$), $d = 16$, $a_0 = 10$. Standard deviation across three seeds at most 1% of the value except where marked. Log loss.

r	$N = 1$		$N = 200$	
	$p = 1$	$p = 0.9$	$p = 1$	$p = 0.9$
0.5	7.6	6.3	2.3	1.9
0.05	16.9	14.8	10.6	9.2
0.001	116.5	23.1	111.4	49.8

5 Unreliable descriptions

Let the learner assume reliability q when the truth is p . Its marginal likelihood is $q \text{ML}_1 + (1 - q) \text{ML}_0$ in terms of the two Gaussian components, so with c the true component and $\pi_c(k)$ the learner’s posterior weight on it after k observations,

$$R_N = \mathbb{E} \left[\frac{1}{2} \sum \text{information gains of the true component} \right] + \mathbb{E} [\log \pi_c(n + N) - \log \pi_c(n)], \quad (2)$$

a Gaussian term as in §4 plus an identification term: the expected rise in the log posterior weight of the truth. Over a long horizon the identification term tends to the cross-entropy $H(p, q)$, so mis-set trust costs $\text{KL}(p \parallel q)$ nats, independent of d ; this is the wrong-prior penalty of universal coding (Cover and Thomas, 2006). We validate the split against the joint Gaussian likelihood and against $\text{KL}(p \parallel q)$: at $p = 0.7$, $q = 0.999$, $d = 16$, the long-horizon excess is 1.46 nats, as predicted.

Long-horizon worth. Combining the split with Proposition 1: as $N \rightarrow \infty$ the description is worth the examples whose information gain is

$$p \frac{d}{2} \log(1/r) - H(p), \quad (3)$$

the reliability-weighted information minus the entropy of not knowing whether the description is right. [Derivation to be written out; verified numerically at one point ($d = 16$, $a_0 = 10$, $r = 0.05$, $p = 0.9$: 9.2 predicted, 9.22 computed).] It is unbounded in precision.

Single-query worth. With no examples, the learner cannot tell whether the description is right, so its predictive is the p -mixture of the two components’ predictives and its regret has a floor of about $(1 - p)$ times the no-description regret. The worth therefore saturates as $r \rightarrow 0$.

Reliability caps the worth of precision. In Table 2, making the description fifty times more precise (r from 0.05 to 0.001) raises the single-query worth from 16.9 to 116.5 examples when it is reliable, but only from 14.8 to 23.1 when it is right nine times in ten. Being wrong one time in a hundred already halves the worth of the most precise description (64.2 at $p = 0.99$). This holds in all four settings we computed ($d \in \{4, 16, 64\}$, $a_0 \in \{10, 100\}$).

The horizon ordering reverses. For the precise, unreliable description the long-horizon worth (49.8) exceeds the single-query worth (23.1), the opposite of §4: with feedback the learner can check the description and then exploit its full precision, whereas for a single prediction it must hedge. The reversal holds in three of our four settings and fails at $d = 16$, $a_0 = 100$, so it is conditional. [Derive the boundary from equation (3) and the single-query floor.]

Trust is cheap to mis-set, except at the extreme. At $d = 16$, $r = 0.001$, $p = 0.7$, the single-query regret is 1.27 nats with calibrated trust, 1.95 at $q = 0.1$, 1.90 at $q = 0.99$, 2.51 when the description is ignored, and 40.5 at $q = 1$. Over a long horizon the excess follows $\text{KL}(p \parallel q)$ and stays below 1.5 nats for any $q \in [0.1, 0.999]$. A Bayes-optimal learner therefore loses little from imperfect trust unless it trusts completely.

Tipping points. A wrong description is abandoned (posterior weight below $1/2$) after a median of one to three examples for $r \leq 0.05$ at every trust level up to $q = 1 - 10^{-6}$; coarse descriptions ($r = 0.5$) take up to 23 examples at $d = 64$. This is far faster than the tens of examples reported for language models (Camassa and Shiller, 2026), which we return to in §7.

6 Do trained networks agree?

[Results pending. The design follows the project’s design record and is fixed; only the numbers are missing.]

Design. We meta-train Transformers on prompts drawn from the process of §3 and compare them with the exact learner on fresh prompts. The prompt follows the prefix layout of Huang and Ge (2025): an optional description row holding m , then example rows, then a question, with two marker columns separating the three kinds of row. The model has no positional information, so the examples have no order, as for the exact learner. Its architecture, optimiser, and learning rate are those of Garg et al. (2022): 12 layers, 8 heads, width 256, Adam at 10^{-4} , fresh prompts at every step. [Output and loss (log loss with a predicted spread, following Genewein et al. (2025), or squared error following Garg et al. (2022)) and scoring (every row or one question) to be fixed.] The description states only m ; its precision and reliability are fixed per model and never stated, so the network learns both from the training distribution, as a language model would from pretraining text. Setting: $d = 5$, $a_0 = 10$, up to 15 examples, half of the training prompts without a description.

Readouts. For each model we report the regret after n examples with and without a description, the description’s worth in examples, and the implied trust: the q at which the exact learner’s prediction is closest to the network’s. Experiments proceed one factor at a time: a reliable precise description; the same model tested on descriptions that are right nine times in ten; a coarse description; seeds; and only then training on unreliable descriptions.

Targets. The exact learner at this setting values the precise description ($r = 0.05$) at 6.7 examples and the coarse one ($r = 0.5$) at 2.2 under log loss (7.4 and 2.7 under squared error); at $p = 0.9$ these fall to 5.0 and 1.8 (4.4 and 2.1).

[Table: network versus exact learner, worth in examples and regret curves, per setting and seed. Figure: regret against n .]

7 Discussion

What the results say. A description’s worth is not one number. For a single answer, the usual way a prompt is used, it is worth more than its information content, and by a factor that grows with the size of the task. Over a session with feedback it is worth exactly its information. Unreliability changes which of these is larger: a precise description that might be wrong is worth little for one answer, because the learner must hedge, and much more once the answers let it check. Precision only pays when the description can be trusted or tested.

Relation to language models. Our learner is Bayes-optimal and our description is a vector; language models are neither. The published record supports two of our qualitative predictions, a tipping point under conflicting examples and lower trust in models whose training text made instructions unreliable (Gupta et al., 2025; Camassa and Shiller, 2026; Bigelow et al., 2025), and says nothing yet about saturation or the horizon. Two disagreements are instructive: language models accumulate evidence sublinearly (Bigelow et al., 2025), and they tip after tens of examples where the exact learner tips after one to three, so a quantitative match would need discounted examples. Whether small meta-trained networks match the exact learner is the question of §6.

Limitations. The description is a vector, not text; the model must weigh it, not parse it. The setting is linear regression with Gaussian inputs, chosen because the optimal learner is exact. The results for unreliable descriptions are computed, with derivations for the long-horizon limit and the trust cost; the boundary of the horizon reversal is not yet derived. [Network results pending.]

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